MINIFLY RESEARCH / DESIGN PROPOSAL / 01

Can answer type transfer to unseen words?

Test whether MiniFly can choose the location from a recalled statement when both the person name and the location word are new. Preserve Round 14's whole-statement writes and question-rarity weighted recall. Change the way recalled candidates are scored.

What Round 14 showed

In one development seed, the word-identity selector scored 99.5% on new locations alone but 49.5% when both names and locations were new. Random-location feedback matched true feedback on the reported QA1 families. This means type-level supervision was sufficient for those tests. It does not show that the learner uses the correct answer for each individual question. The shared-word qfound stress case scored 58%, which motivates a separate recall diagnostic.

Preserve a small amount of context with each candidate

When a sentence ends, use the existing byte n-gram address and fixed local write rule. Store every word candidate with a bounded descriptor of its adjacent tokens. A starting descriptor is the nearest two tokens on each side, with a candidate placeholder and a fixed boundary symbol. Exclude the candidate's own bytes from the context score. Preserve left and right adjacency, with no absolute slot number, named semantic role, answer vocabulary filter or learned write gate.

The original word bag cannot recover those descriptors by itself. Extend the existing fixed-address payload to carry candidate and descriptor associations together, using a capped candidate budget and reporting the additional bytes and collisions. A recalled candidate must obtain its descriptor from that payload. No scan of saved sentences or evaluator reconstruction of candidate context is available to the learner.

Local selection rule

After the answer arrives during teaching, count a hit for each recalled candidate equal to that word and a trial for every recalled candidate. Use the Round 14 smoothed base-rate rule for word identity and the same rule for a contextual arm with fixed hashed descriptor features. Average the active feature scores with smoothing toward the training base rate. Freeze feature sizes, smoothing and tie rules after engineering checks. During evaluation, freeze selection and question-rarity statistics. Story memory continues to write complete statements.

This test retains sentence segmentation, word splitting, an explicit query cue and whole-word answers. It measures word selection and transfer across lexical substitutions. Byte generation and end-of-answer behavior will require a separate test.

Paired tests and causal controls

Compare selectors on the same recalled candidates

Primary comparison. Context-only versus Round 14 word-identity scoring, both using the same augmented store, candidate budget and teaching stream. Replay the original unmodified Round 14 path on engineering worlds to measure any effect from the payload change itself.

Feedback control. Run both selectors with true feedback and with location words sampled independently of the queried event, preserving the training location frequencies and feedback schedule. Include a no-feedback reference with a predeclared tie rule. Report target availability in the candidate set under each feedback arm. Feedback never changes the story writes.

Context control. At evaluation, permute recalled descriptors among candidates within each event while preserving the candidate words and scores' feature budget. Use fixed permutation seeds. A contextual gain that disappears under permutation depends on the binding between words and their contexts.

Read diagnostic. Measure whether the correct answer is in the actual recalled set before selection. For a separate evaluator-only analysis, score the selectors on words and descriptors from the true supporting statement. This distinguishes bad recall from bad selection. Keep supporting-fact metadata outside training and deployment.

Sealed QA1 worlds

Use 12 new story batches and three fixed hash seeds. Preserve the same latent events, queries and answers across paired renderings. Include familiar wording, new names only, new locations only, and both jointly new. Assign new names and locations independently with disjoint identities from training. For the joint condition, report two prespecified panels: familiar contexts and held-out statement templates. Seal templates, worlds and hash seeds before scoring.

Add a destination-versus-distractor panel with two location-like words in the same statement. Counterbalance which comes first and verify the intended destination with the generator. Report qfound separately as a stress test for the retrieval stage. Neither panel is included in the main QA1 score.

Do not feed back gold answers at evaluation. Report exact selected-word accuracy, candidate-set recall, selector accuracy conditional on target availability, tie rate, store occupancy, collisions and bytes per event. Estimate paired intervals over story batches after averaging the hash seeds within each batch.

Decisions and follow-up

Proposed primary gate

On jointly new names and locations in familiar contexts, require at least 85% exact selected-word accuracy and a paired gain of at least 20 percentage points over word-identity selection, with a 95% interval excluding zero. Require at least 90% target availability and no more than a 3-point mean loss on familiar wording. Report the held-out-template panel separately. Reserve a claim of template transfer for a panel that independently meets the same gate.

If permuting descriptors leaves the gain intact, inspect candidate order, tie behavior and leaked identity information before claiming contextual learning. If contextual selection works on the true supporting event but fails on recalled candidates, prioritize retrieval. If it works only when there is one location-like candidate, the supported conclusion is answer-type selection. Choosing the correct destination among multiple locations remains open.

What the feedback comparison can establish

A tie between true and random-location feedback on the primary test would again support type learning. Use the two-location panel to ask whether true feedback offers a measurable advantage when the candidate words share the same type. Report that contrast separately. This QA1 experiment alone supplies no evidence for multistep reasoning or long-term retention across stories.

Engineering and next experiments

Cycle 1: verify candidate-context binding, equal feature budgets, strictly delayed feedback and freeze hashes. Cycle 2: run complete development stories for all controls, audit coverage and descriptor permutation, and verify that the augmented payload preserves baseline recall. Cycle 3: verify fresh-process checkpoint recovery, deterministic output and corrupt checkpoint rejection. Seal code, stories, thresholds and reducer after these checks.

A passing selector can be tested next with byte-by-byte output and model-generated termination. Keep QA2 as a separate comparison: whole-statement storage and two fixed causal recalls, with a controlled test of whether the learned selector can replace positional roles. Hold question-rarity weighting and segmentation fixed until the selection result is interpretable.

Source and status

Basis: MiniFly Round 14 scaffold-removal memo, development results on 200 questions per family at seed 1. The proposed representation, run size and gates are design choices. No new experiments were launched.